Gen AI Architect Program

Assignment 3 - Streamlit Student Grade App | learn. build. operate

1. What you will build

Build a small web app using the **Streamlit** framework that turns a mark into a letter grade. Instead of running in the terminal, your app opens in the browser with a simple user interface: the user types or selects a mark (a number between 0 and 100) in an input widget, and the app displays the matching grade on the page. You will set up a clean Python environment, install Streamlit, write your code in a **.py** file, run the app, and confirm it shows the correct grade.

Difficulty: Beginner to Medium · Tools: Python, Streamlit

2. Grading scale to implement

Your app must convert a mark into a grade using exactly this scale:

Mark range	Grade
90 - 100	A
80 - 89	B
70 - 79	C
60 - 69	D
Below 60	E

Note: the boundaries are inclusive - a mark of exactly 90 is an A, exactly 80 is a B, and so on.

3. Requirements

- Use the **Streamlit** framework to build the user interface. Pure Python for the grading logic.
- Provide an input widget where the user can enter or choose a mark.
- Display the entered mark and the resulting grade clearly on the page.
- Handle the full 0-100 range and cover every grade band above.
- Give the app a title or heading so it looks finished.
- Keep all your code in a single file named `grade_app.py`.

Think about edge cases

What should happen if someone enters a number above 100, below 0, or leaves the field empty? Decide how your app should respond, and make sure it shows a friendly message instead of crashing.

4. Step-by-step setup

Follow these steps in order. Commands are shown for both macOS/Linux and Windows where they differ.

Step 1 - Create a project folder

```
mkdir grade-app
cd grade-app
```

Step 2 - Create a virtual environment (venv)

A virtual environment keeps this project isolated from the rest of your system.

```
python -m venv venv
# (on some systems use 'python3' instead of 'python')
```

Step 3 - Activate the virtual environment

```
# macOS / Linux:
source venv/bin/activate

# Windows (PowerShell):
venv\Scripts\Activate.ps1

# Windows (Command Prompt):
venv\Scripts\activate.bat
```

When activated, you will see `(venv)` at the start of your terminal prompt.

Step 4 - Install Streamlit

Install the Streamlit framework inside the activated venv.

```
pip install streamlit
```

Step 5 - Run your app

Streamlit apps are started with `streamlit run`, not `python`.

```
streamlit run file_name.py
```

The app opens automatically in your browser at `http://localhost:8501`. Test it with several marks to confirm each grade band works.

Step 6 - Stop and deactivate when finished

Press `Ctrl + C` in the terminal to stop the app, then deactivate the venv.

```
deactivate
```

5. How the app should behave

Your layout and wording can differ, but a typical interaction should look like this:

```
[ Enter your mark (0-100): 85 ]

Mark: 85 -> Grade: B
```

6. What to submit

- Your `grade_app.py` file.
- A short screenshot or screen recording of the app running in the browser with at least three different marks.
- Two or three sentences explaining how you handled invalid input.
- Upload the code to GitHub and send us the repository link.

Reminder: this is a solo task. Build the logic and the UI yourself - the steps above are your setup roadmap, not the answer. Good luck!